

Human Review: separable-dual wording loophole

Packet identifier: 2509.00722_separable_free_space_dual_embedding

Original automatic proof: solution_packet.pdf

Checked date: 14 June 2026

Status: Manually checked.

Verdict: Rejected as a substantive solution to the intended problem. This is a wording loophole rather than a mathematical resolution.

Human Comments

The proposed solution in `solution_packet.pdf` should not be accepted as a full solution of Problem 3.12 in Smith’s paper. The argument it gives is mathematically correct as a literal Banach-space observation: if Y is separable, then a countable norming family in Y^* gives an isometric embedding

$$Y \hookrightarrow \ell_\infty = (\ell_1)^*.$$

Since $\mathcal{F}(M)$ is separable whenever M is separable, this gives $\mathcal{F}(M) \hookrightarrow (\ell_1)^*$ for every separable metric space M .

However, this cannot be the intended mathematical content of Problem 3.12. The paper explicitly defines $X \hookrightarrow Y$ to mean linear isomorphism to a subspace, so the automatic packet is exploiting a literal reading of the notation. But the surrounding context makes that reading vacuous. Theorem 1.3 constructs countable complete discrete metric spaces M for which $\mathcal{F}(M)$ fails the bounded approximation property and consequently is not isomorphic to any dual Banach space. Immediately after related remarks about complete separable plu spaces, the author says that G. Lancien suggested Problem 3.12 as something Theorem 1.3 leaves open.

Under the packet’s reading, none of the hypotheses or preceding discussion matter: every separable Banach space embeds into the dual of the separable space ℓ_1 . Thus the question would have a one-line affirmative answer for all separable M , independent of plu, completeness, the Radon–Nikodym property, approximation properties, or the issue of being isomorphic to a dual. This is strong evidence that the intended question involved a stronger duality requirement, or that the printed problem is missing a qualifier.

Possible intended variants include asking whether $\mathcal{F}(M)$ is itself isomorphic to a dual space with separable predual, whether it embeds into a separable dual in a weak-star meaningful way, or whether some related complemented or canonical dual structure exists. The automatic packet does not address any such substantive variant.

Short Version

The $\ell_\infty = (\ell_1)^*$ embedding is correct, but it makes Problem 3.12 completely trivial for every separable Banach space. The source context is about whether certain Lipschitz-free spaces are, or meaningfully sit in, dual spaces. Therefore this packet should be treated as a wording observation, not as a mathematical solution of the intended problem.

Review Outcome

Reject this candidate as a successful full solution. At most, archive it as a cautionary example: the automatic system found a literal loophole in the printed wording, but the surrounding paper makes

clear that the intended question is a nontrivial duality question not answered by the universal ℓ_∞ embedding.